SUBJECT
Bt501' (important past papers)

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Bt501 solved important past papers for mid term by vu medical zone

What is CFTR? 2

CFTR is an abbreviation of Cystic fibrosis transmembrane conductance regulator. **CFTR** is a membrane protein and chloride channel in vertebrates that is encoded by the **CFTR** gene. The **CFTR** gene codes for an ABC transporter-class ion channel protein that conducts chloride ions across epithelial cell membranes.

What are techniques in rDNA Technology? 2

There are many techniques used in rDNA technology these are:

1) Gel electrophoresis
2) Cloning libraries
3) Restriction enzyme mapping
4) PCR
5) Nucleic Acid Hybridization
6) DNA Microarrays

Differnce b/w allogenic and autologus stem cells? 3

Allogenic stem cells: Autologous Stem Cells: Sources of the patient's own stem cells (autologous) are either the cells from patient's own body or his or her cord blood. For autologous transplants physicians now usually collect stem cells from the peripheral blood rather than the marrow	Autologous stem cells: Allogenic Stem Cells: Sources of stem cells from another donor (allogeneic). These are primarily relatives (familial-allogeneic) or completely unrelated donors (unrelated-allogeneic).The stem cells in this situation are extracted from either the donor's body or cord blood

Write a short note on Alzheimer dsisease.? 3

Alzheimer diseases (AD):

is a panethnic, genetically heterogeneous disease; less than 5% of patients have early-onset familial disease, 15% to 25% have late-onset familial disease, and 75% have sporadic disease. Approximately 10% of familial AD exhibits autosomal dominant inheritance; the remainder exhibits multifactorial inheritance. Current evidence suggests that defects of β-amyloid precursor protein metabolism cause the neuronal dysfunction and death observed with AD. Consistent with this hypothesis, mutations associated with early-onset autosomal dominant AD have been identified in the β-amyloid precursor protein gene (APP), the presenilin 1 gene (PSEN1), and the presenilin 2 gene (PSEN2).

How Diabetes act as polygenic disorder.? 5

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Polygenic disease: A genetic disorder that is caused by the combined action of more than one gene. Examples of polygenic conditions include hypertension, coronary heart disease, and diabetes. Because such disorders depend on the simultaneous presence of several genes, they are not inherited as simply as are single-gene diseases.

Examples of polygenic conditions include hypertension, coronary heart disease, and diabetes. Because such disorders depend on the simultaneous presence of several genes, they are not inherited as simply as are singlegene diseases. Genetic disorders may also be complex, multifactorial, or polygenic, meaning they are likely associated with the effects of multiple genes in combination with lifestyles and environmental factors. Multifactorial disorders include heart disease and diabetes. **Diabetes mellitus** is the most common endocrine disease, consuming almost 15% of annual health care expenditures in the United States. The focus of this review is on type II (non-insulin- dependent) diabetes mellitus, which accounts for approximately 90% of diabetic patients. The current understanding of the pathogenesis of type II diabetes is multifaceted. Complex defects in both insulin action and insulin secretion produce the metabolic derangements responsible for the disease. The pathophysiology is the result of a combination of polygene defects and environmental factors. The search for the responsible gene or genes is described and several candidate genes are discussed. Diabetogenes are addressed in the context of type II diabetes representing a prototype for the identification of the genetic basis of common diseases.

What is Human genome project.? 5

International Human Genome Sequencing Consortium: ▪Proposed 1985, endorsed in 1988. ▪20 governmental groups. ▪"Public project."	Craig Venter & Celera Genomics: ▪ Founded 1998. ▪ Sequence in 3 years. ▪ Technology: automation, computers. ▪ Had access to public project's data.	International Human Genome Sequencing Consortium Approach was conservative and methodical. Had to wait for technology. First produced a clone-based physical map of the genome that would serve as a scaffold for the later sequence data: Broke genome into chunks of DNA whose position on chromosome was known from maps, clone into bacteria using BACs. Digest BAC-inserted clonal chunks of DNA into small fragments. Sequence small fragments. Stitch together BAC clones to assemble sequence. Assemble genome sequence from BAC clone sequences, using clone-based physical map.

The **Human Genome Project** (**HGP**) was an international scientific research project with the goal of determining the sequence of nucleotide base pairs that make up human DNA, and of identifying and mapping all of the genes of the human genome from both a physical and a functional standpoint. It remains the world's largest collaborative biological project. After the idea was picked up in 1984 by the US government when the planning started, the project formally launched in 1990 and was declared complete on April 14, 2003. Funding came from the US government through the National Institutes

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of Health (NIH) as well as numerous other groups from around the world. A parallel project was conducted outside government by the Celera Corporation, or Celera Genomics, which was formally launched in 1998. Most of the government-sponsored sequencing was performed in twenty universities and research centers in the United States, the United Kingdom, Japan, France, Germany and China. The Human Genome Project originally aimed to map the nucleotides contained in a human haploid reference genome (more than three billion). The "genome" of any given individual is unique; mapping the "human genome" involved sequencing a small number of individuals and then assembling these together to get a complete sequence for each chromosome. Therefore, the finished human genome is a mosaic, not representing any one individual.

How real time Pcr is used for immunoassay infectious agent?

The Polymerase Chain Reaction (PCR) is a process for the amplification of specific fragments of DNA. Real-Time PCR a specialized technique that allows a PCR reaction to be visualized “in real time” as the reaction progresses. As we will see, Real-Time PCR allows us to measure minute amounts of DNA sequences in a sample. This method also allows a rapid assessment of the presence of antibiotic resistance genes or gene mutations. Although this genetic approach is not always predictive of phenotypic resistances, in specific situations it may help to optimize the therapeutic management of patients. Finally, an approach combining the detection of pathogens, their mechanisms of antibiotic resistance, their virulence factors and bacterial load in clinical samples could lead to profound changes in the care of these infected patients. In recent years, quantitative real-time PCR tests have been extensively developed in clinical microbiology laboratories for routine diagnosis of infectious diseases, particularly bacterial diseases. This molecular tool is well-suited for the rapid detection of bacteria directly in clinical specimens, allowing early, sensitive and specific laboratory confirmation of related diseases. It is particularly suitable for the diagnosis of infections caused by fastidious growth species, and the number of these pathogens has increased recently .

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